

Lab 6: ARCH effects, GARCH fitting and Value-at-Risk

Volatility clustering, Engle LM, GARCH(1, 1) and forecasting

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Goals

- Detect volatility clustering visually and via the Engle LM test.
- Estimate ARCH(q) and GARCH(p, q) models by maximum likelihood.
- Validate the fit through standardised-residual diagnostics.
- Produce conditional volatility forecasts and a one-day Value-at-Risk.

Setup

- Python 3 with `numpy`, `pandas`, `matplotlib`, `statsmodels` and `arch`.
- `numpy.random.seed(42)`.
- Dataset: `data/synthetic_returns.csv` – $T = 1500$ synthetic daily returns generated from a GARCH(1, 1) with $(\alpha_0, \alpha_1, \beta_1) = (0.05, 0.10, 0.85)$ and Gaussian innovations.

Warning

Install the `arch` package (Kevin Sheppard) once via `pip install arch`. The main entry point is `from arch import arch_model`.

Exercise 1 – Visualising volatility clustering

- (1) Load `data/synthetic_returns.csv`. Plot the return series r_t versus t . Comment: do you observe quiet periods alternating with turbulent ones?
- (2) Compute and plot the sample ACF of r_t up to lag 20. The returns should look (almost) uncorrelated.
- (3) Plot the sample ACF of r_t^2 up to lag 20. Compare with the previous plot and comment in 3 lines.

Exercise 2 – Engle’s LM test

- (1) Use `statsmodels.stats.diagnostic.het_arch` on r_t at lags 5, 10, 20. Build a small table with columns: lag, LM statistic, $\chi_{q,0.95}^2$ critical value, p -value, decision.
- (2) Conclude: is there evidence of ARCH effects in the data? Why is this consistent with what the ACF of r_t^2 already showed?

Exercise 3 – ARCH(q) fitting

Use `arch_model(returns, vol="ARCH", p=q, mean="Constant")`. Note: in the `arch` package the “`p`” parameter is the ARCH order.

- (1) Fit ARCH(1), ARCH(3) and ARCH(5). Report $\hat{\alpha}_0$, the $\hat{\alpha}_i$, the log-likelihood and the AIC for each.
- (2) Compute $\sum_i \hat{\alpha}_i$ for each. Is it less than 1 (covariance-stationarity)?
- (3) Why does ARCH(5) have only a marginal AIC gain over ARCH(3)? Hint: the true model has volatility memory not directly captured by simple lagged squared shocks.

Exercise 4 – GARCH(1, 1) fitting

- (1) Fit GARCH(1, 1) with constant mean and Gaussian innovations:

$$r_t = \mu + \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

Report $\hat{\mu}$, $\hat{\alpha}_0$, $\hat{\alpha}_1$, $\hat{\beta}_1$, the log-likelihood and the AIC.

- (2) Compute $\hat{\alpha}_1 + \hat{\beta}_1$ and the implied unconditional variance $\hat{\sigma}^2 = \hat{\alpha}_0 / (1 - \hat{\alpha}_1 - \hat{\beta}_1)$. Compare with the empirical sample variance.
- (3) Compare AIC: GARCH(1, 1) vs. ARCH(5). Comment: is the parsimonious GARCH(1, 1) preferred?

Exercise 5 – Standardised-residual diagnostics

Compute $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ from the GARCH(1, 1) fit.

- (1) Plot the ACF of $\hat{z}_t$ and of $\hat{z}_t^2$ up to lag 20.
- (2) Run Engle’s LM test on $\hat{z}_t^2$ at the same lags as Exercise 2. The model is well-specified if it does *not* reject anymore.
- (3) Plot a QQ-plot of $\hat{z}_t$ vs. $\mathcal{N}(0, 1)$. Comment on the marginal fit.

Exercise 6 – Volatility forecast and VaR

- (1) On the GARCH(1, 1) fit, compute the one-step forecast $\hat{\sigma}_{T+1}$ and the iterated h -step forecasts $\hat{\sigma}_{T+h}^2$ for $h = 1, \dots, 30$. Plot $\hat{\sigma}_{T+h}^2$ together with the unconditional variance $\hat{\sigma}^2$.
- (2) Verify numerically the mean-reversion formula

$$\hat{\sigma}_{T+h}^2 = \hat{\sigma}^2 + (\hat{\alpha}_1 + \hat{\beta}_1)^{h-1} (\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2).$$

- (3) Compute the one-day Value-at-Risk at level $\alpha \in \{1\%, 5\%\}$:

$$\text{VaR}_{T+1}(\alpha) = -(\hat{\mu} + z_\alpha \hat{\sigma}_{T+1}).$$

Compare with the unconditional VaR using the sample σ . Which one is tighter today?

Warning

$z_\alpha = \Phi^{-1}(\alpha)$ is computed via `scipy.stats.norm.ppf(0.01) = -2.326` and `norm.ppf(0.05) = -1.645`.